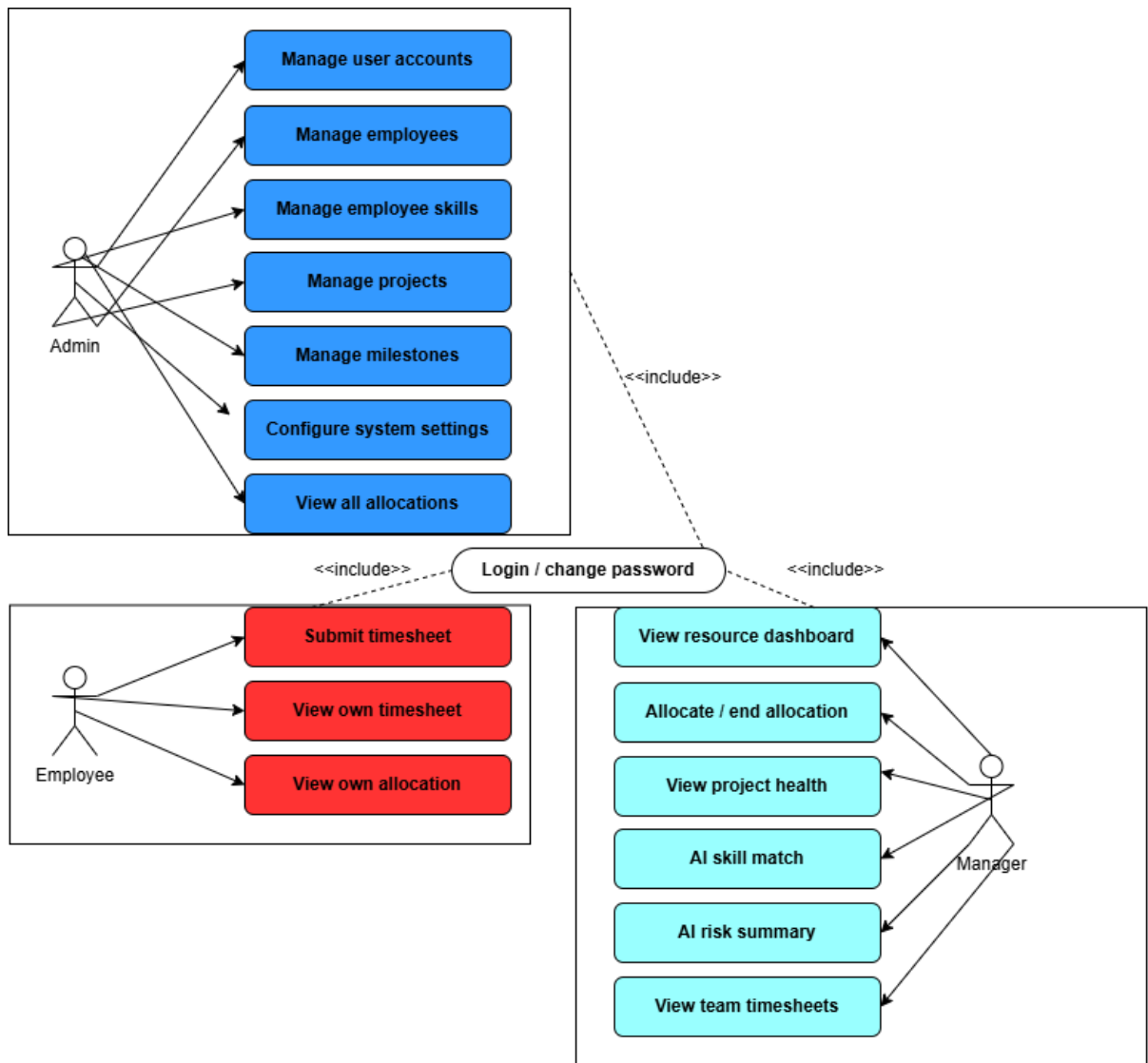


Introduction - Project and Resource Management (PRM)

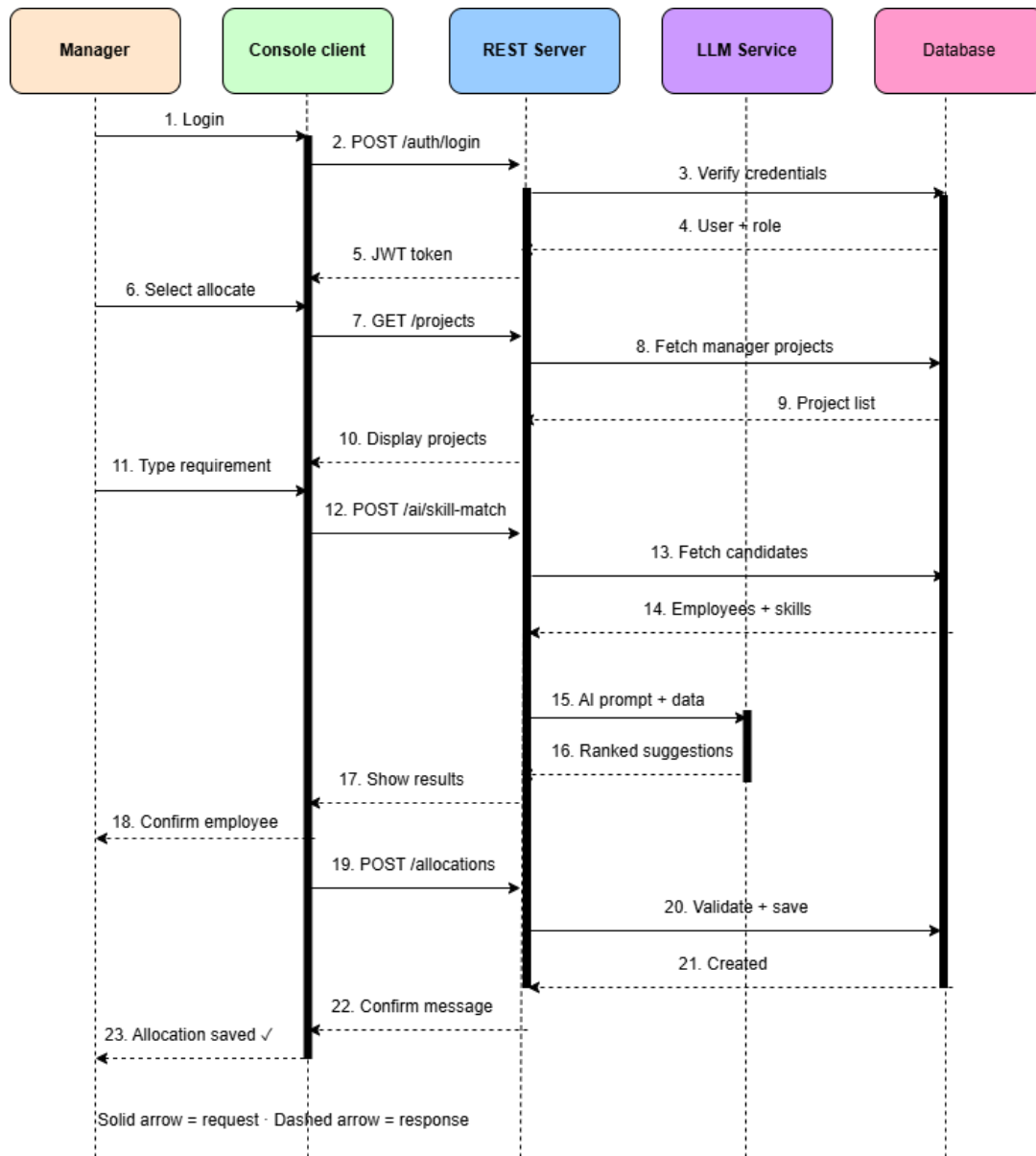
The Project and Resource Management (PRM) Tool is a console-based client-server application designed to streamline resource planning, project management, and workforce utilization within service-based IT organizations. The system provides a centralized platform for managing employees, projects, skills, allocations, milestones, and timesheets while eliminating the inefficiencies associated with spreadsheet-driven processes. In addition to core project management capabilities, the application incorporates Artificial Intelligence (AI) features such as skill-based resource recommendation and project risk analysis to support informed decision-making.

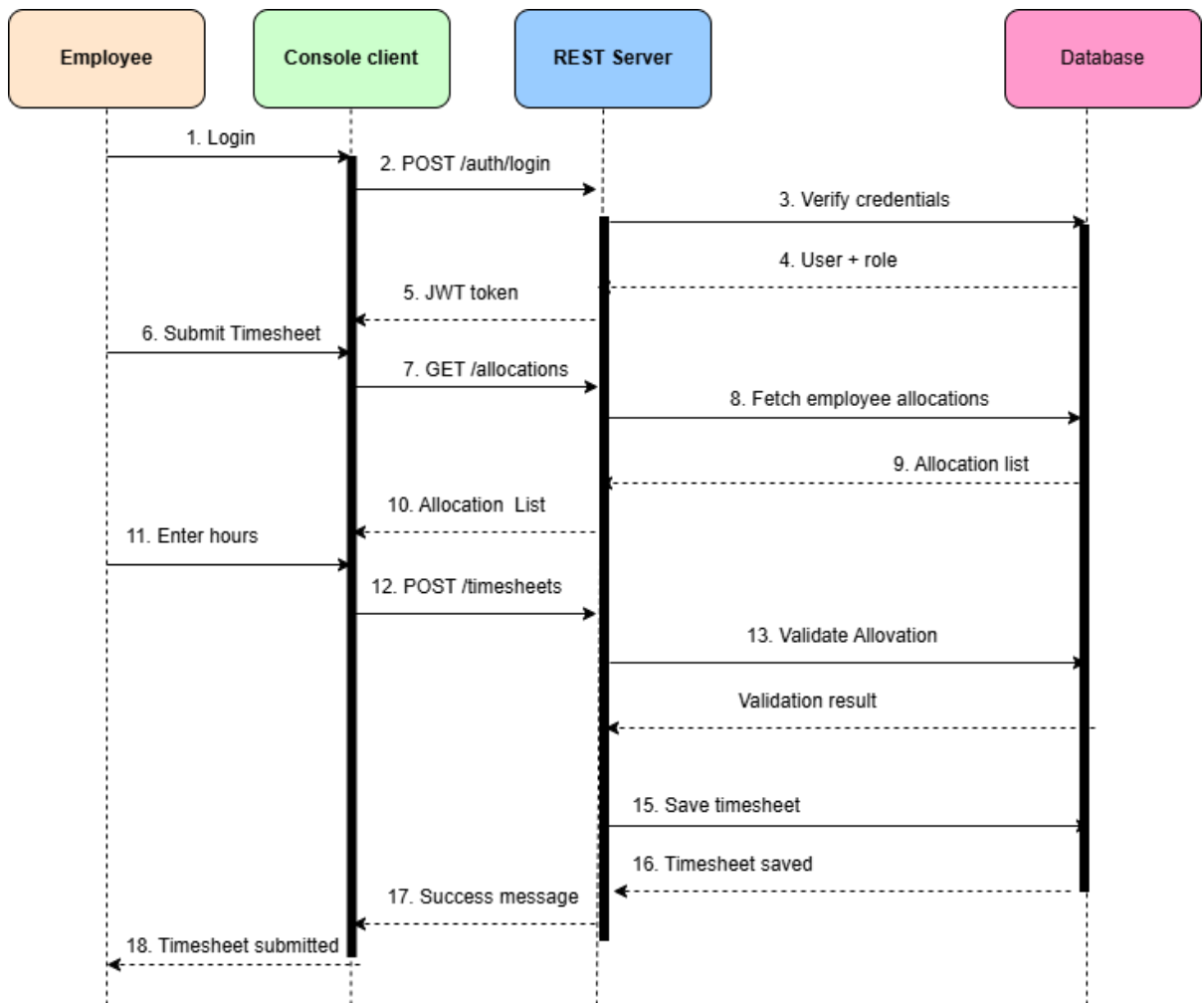
This document presents the high-level system design of the PRM Tool through a collection of UML and database design artifacts, including Use Case Diagrams, Sequence Diagrams, Class Diagrams, and Entity Relationship (ER) Diagrams. These diagrams collectively describe the system's functional requirements, user interactions, application architecture, object relationships, and database structure, providing a clear blueprint for implementation and future maintenance. The design serves as the foundation for developing a scalable, maintainable, and extensible solution that adheres to software engineering best practices and object-oriented design principles.

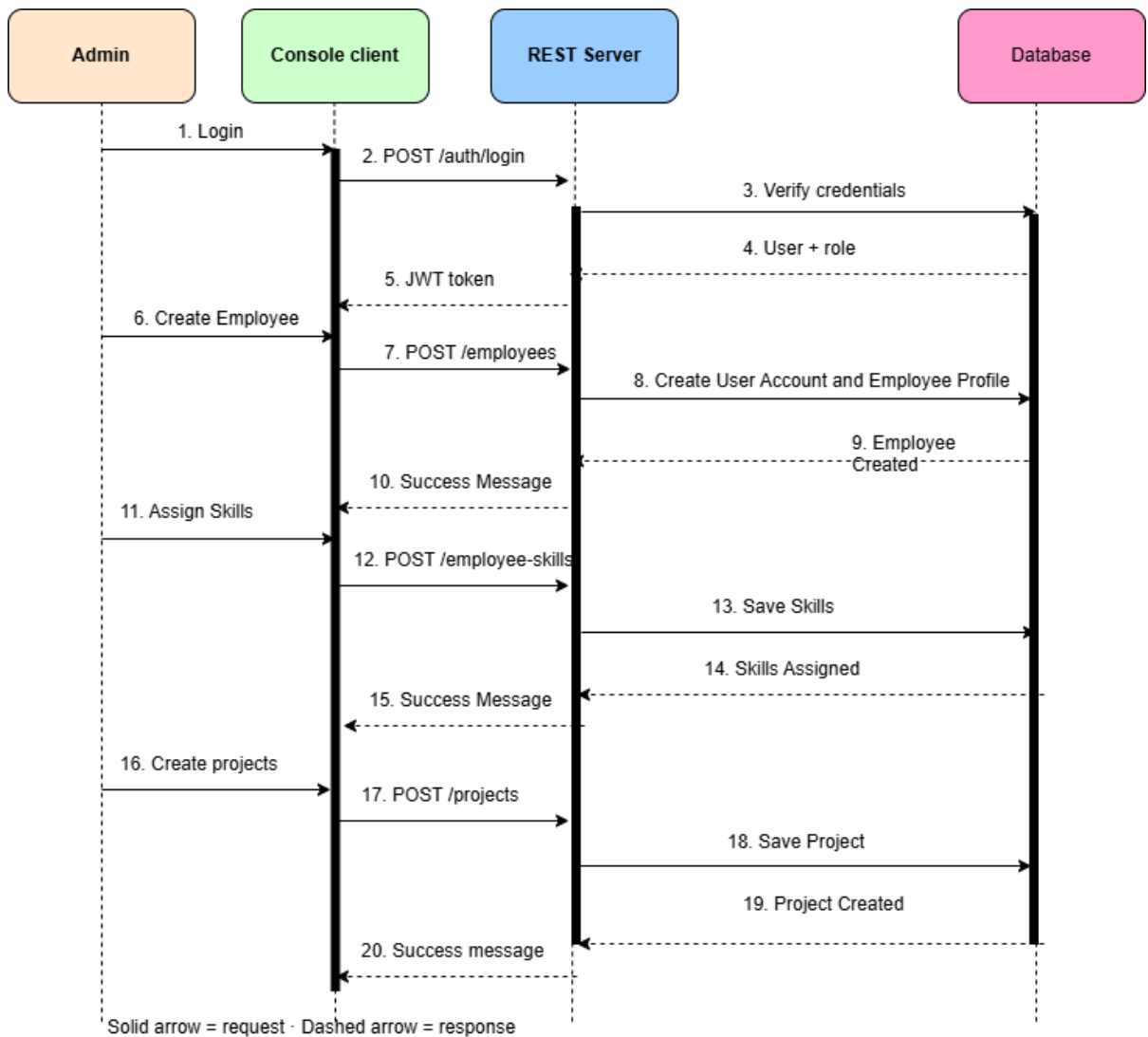
1. Usecase Diagram:



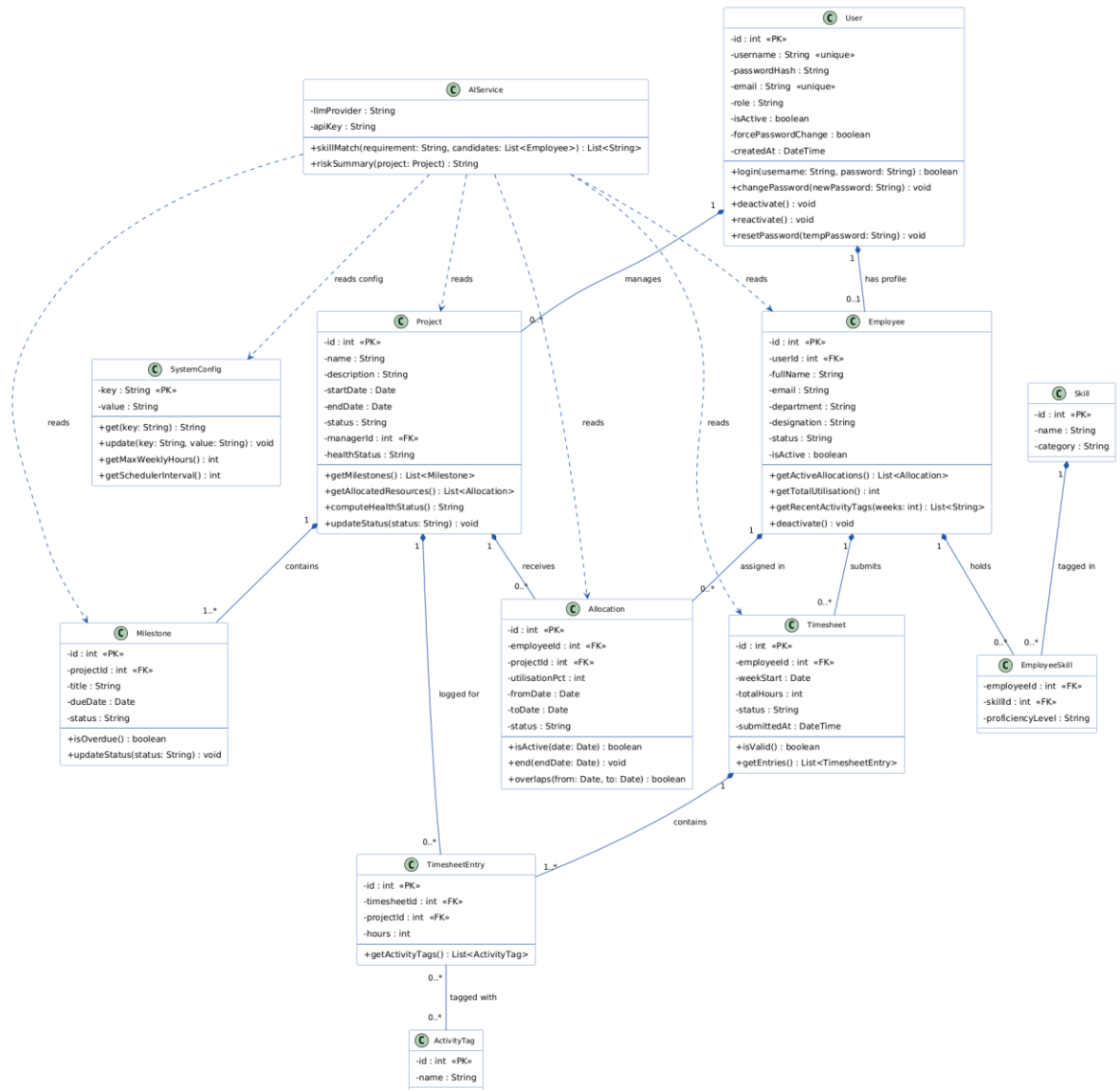
2. Sequence Diagram:







3. Class Diagram:



4. ER Diagram:

